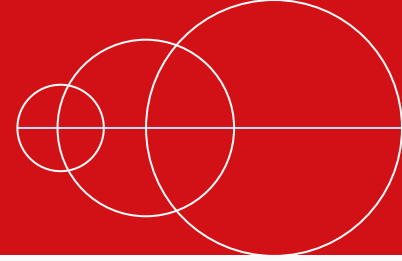


Active User Behavior *and segmentation*

RevoU x Telkom Indonesia



Disclaimer

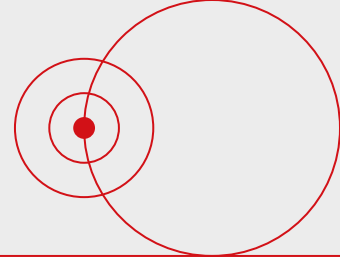


The data provided is a representation of the original data which amount has been adjusted.

The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company.



Clustering Overview



Objective: segment customers based on engagement + value to guide personalization, churn prevention, and upsell.

Model: Rule-based **RFM (SQL)**

- **R (Recency):** days since last activity
- **F (Frequency):** # active days
- **M (Monetary):** total charged revenue (`is_charge = TRUE`)
Supporting: `channel_diversity`, `total_events`, `charged_events`, `tenure_days`

Segment output (N=20):

- High 6 (30%), Medium 7 (35%), Low 7 (35%)



Current Usage Distribution

MONTH YEAR

Key pattern in the dataset

- Recency & Frequency show limited variation across users (high overall activity consistency).
- Channel usage is broadly multi-channel.

Implication

- Segmentation is most strongly differentiated by **Monetary** (charged revenue) and event volume.



Segment Characteristics

MONTH YEAR

Segment	RFM characteristics (typical)	What it means (behavior/value)
High	R high, F high, M high	High revenue
Medium	R high, F high, M medium	Medium revenue
Low	R medium, F medium, M low	Low revenue



Segment Distribution

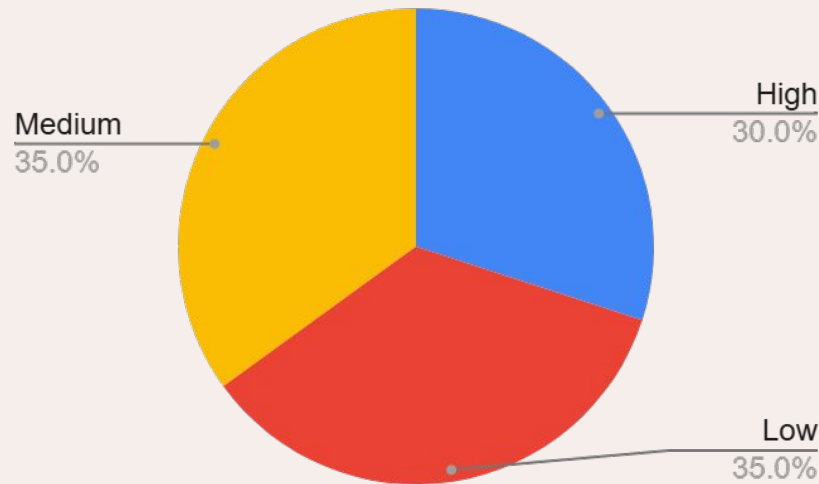


Segment split (N=20)

- High: 30% (6)
- Medium: 35% (7)
- Low: 35% (7)

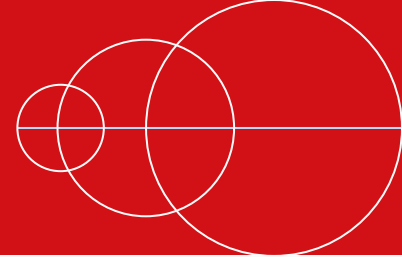
Key insights

- The customer base is **balanced**, with **70%** sitting in **Medium + Low** → biggest growth lever is upgrading these groups.
- **High segment (30%)** likely contributes a disproportionate share of revenue → retention and expansion here has the highest ROI.
- Medium and Low are equal in size → you can run **two different playbooks**: “upsell” for Medium, “reactivation/nurture” for Low.



Customer Segment Distribution

Recommendations



High (6) — Protect & Expand

- Priority support + regular check-ins
- Upsell to higher-volume workflows (bundles, automation, multi-channel)
- Co-plan Ramadan / seasonal campaigns to maximize billable volume

Medium (7) — Upsell Pipeline

- Identify 2–3 best use cases per customer to increase chargeable events
- Offer adoption goals + playbooks (campaign cadence, timing optimization)
- Incentives to scale usage (credits/bundles)

Low (7) — Low-Cost Growth / Nurture

- Light-touch enablement: onboarding refresh, templates, FAQs
- Start with low-effort channels/use cases and monitor improvement
- Deprioritize high-cost support until usage increases

Monitoring (simple rule)

- Track weekly **charged events + revenue**; flag sudden drops to prevent slipping from High/Medium to Low.



Most Profitable Segment

MONTH YEAR

Key insight

- **High is the most profitable segment** by both **total** and **average** revenue:
 - High contributes **~4.6×** the total revenue of Medium (**19.47M vs 4.26M**)
 - High contributes **~8.8×** the total revenue of Low (**19.47M vs 2.22M**)
 - Per customer, High generates **~5.3×** Medium and **~10.2×** Low.

Recommendations

- **High:** Protect & expand (priority support, dedicated check-ins, upsell to higher-volume workflows).
- **Medium:** Best upsell pipeline (adoption goals + bundles/automation to move Medium → High).
- **Low:** Low-cost nurture (templates, onboarding refresh, light-touch campaigns; monitor for uplift).

Segment	Total Revenue	Average Revenue
High (~75% of revenue)	19,465,030	3,244,171
Low	2,218,430	316,918
Medium	4,262,960	608,994



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30-Day Trend Insight (Last 30D vs Previous 30D)

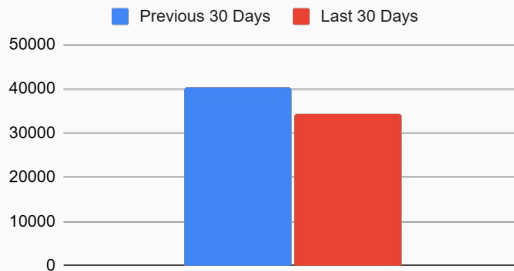
Key KPI movement (All 20 customers)

- **Total events:** 34,349 vs 40,472 → **-15.1%**
- **Charged/successful events:** 27,339 vs 31,890 → **-14.3%**
- **Charged revenue:** 8,221,400 vs 8,603,920 → **-4.45%**

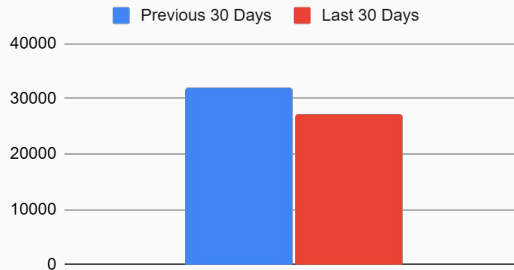
Core takeaway

- Overall usage declined ~15%, but revenue only declined ~4.5% → **revenue is more resilient than activity** (likely due to higher revenue per successful/charged event or mix shift).
- This period also aligns with major holidays: **Ramadan began 1 March 2025** and **Idul Fitri fell on 31 March 2025** (Indonesia), which usually amplifies promotional activity.

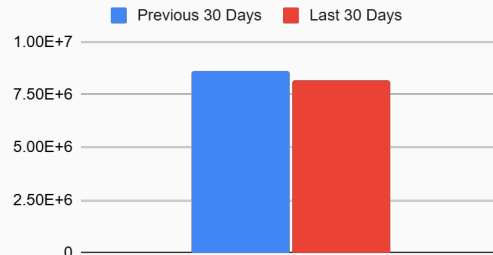
Total Transactions



Charged Messages



Total Revenue



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Clients Prone to Churn

User	Charged Events Last 30D	Charged Events Prev 30D	Change Percentage
CV Niaga Indo	325	438	-25.80%
CV Mart Indo	306	414	-26.09%
CV Hotel Nusantara	310	437	-29.06%
PT Travel Nusantara	304	433	-29.79%

User	Revenue Last 30D	Revenue Prev 30D	Change Percentage
PT Logistik Mandiri	390,760.00	452,060.00	-13.56%
PT Travel Nusantara	103,110.00	120,830.00	-14.67%
CV Hotel Nusantara	87,660.00	109,730.00	-20.11%
CV Mart Indo	88,720.00	111,740.00	-20.60%

Early-risk watchlist (largest usage drops; near -30% threshold)

- **PT Travel Nusantara:** charged ratio **-29.8%**
- **CV Hotel Nusantara:** charged ratio **-29.1%**
- **CV Mart Indo:** charged ratio **-26.1%**
- **CV Niaga Indo:** charged ratio **-25.8%**

Revenue decline watchlist (largest value drops)

- **CV Mart Indo:** **-20.6%**
- **CV Hotel Nusantara:** **-20.1%**
- **PT Travel Nusantara:** **-14.7%**
- **PT Logistik Mandiri:** **-13.6%**

Insights and Recommendations



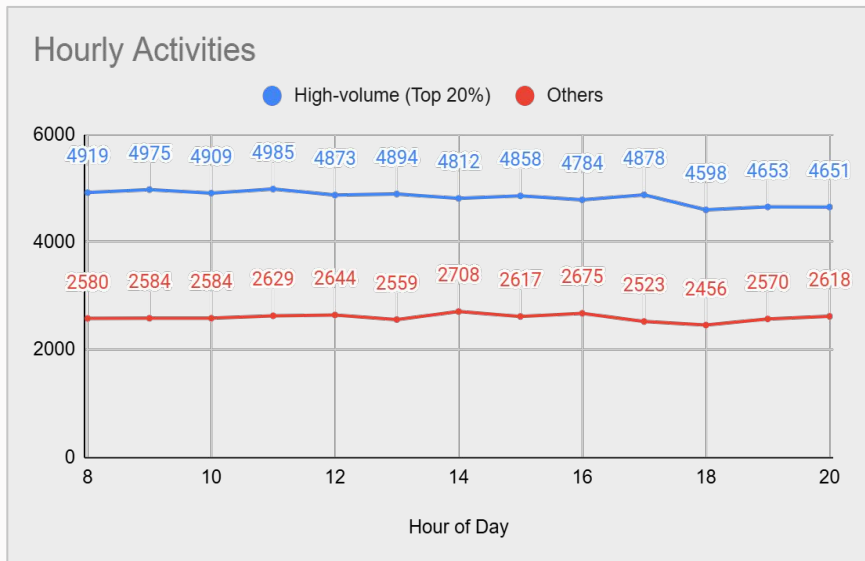
Insights

- **Usage is declining faster than revenue:**
 - Total events -15.1%
 - Charged/successful events -14.3%
 - Charged revenue -4.45%
 - Revenue is relatively resilient vs activity (likely higher revenue per charged event / mix shift).
- **Risk is concentrated in a few customers:** biggest drops are clustered in a small watchlist.
- **Two risk lenses:**
 - **Usage drop %** (early warning)
 - **Revenue drop %** (business impact now)

Recommendations

- **Prioritize the watchlist immediately (highest ROI)**
Focus on customers with **≥25% charged-events drop** to prevent deeper declines.
- **Run a quick diagnosis (pick the right fix)**
 - If decline is **lower demand/campaigns** → propose new use-cases + automation journeys to recover usage.
 - If decline is **delivery/channel friction** → troubleshoot message status issues and optimize channel + timing.
- **Set a short recovery target (2–4 weeks)**
Aim to bring charged-events drop back from **-25%~ -30%** toward **0%** (baseline).
- **Monitoring rule going forward**
Weekly alert if **charged_events % change ≤ -25%** or **revenue % change ≤ -15%**.

Hourly Activity Pattern



What the chart shows

- Both groups concentrate activity during **business hours (08:00–17:00)**, then dip after **18:00**.
- High-volume (Top 20%)** consistently generates **~1.8x** more activity than Others across all hours.

Peak hours (clear differences)

- High-volume peak: 11:00** with **4,985** charged events
 - Strong window: **09:00–13:00** (all near ~4.9k)
- Others peak: 14:00** with **2,708** charged events
 - Secondary peak: **16:00** with **2,675**

Notable dip

- Both groups hit the lowest point around **18:00**
 - High-volume: **4,598**
 - Others: **2,456**

Insights and Recommendations

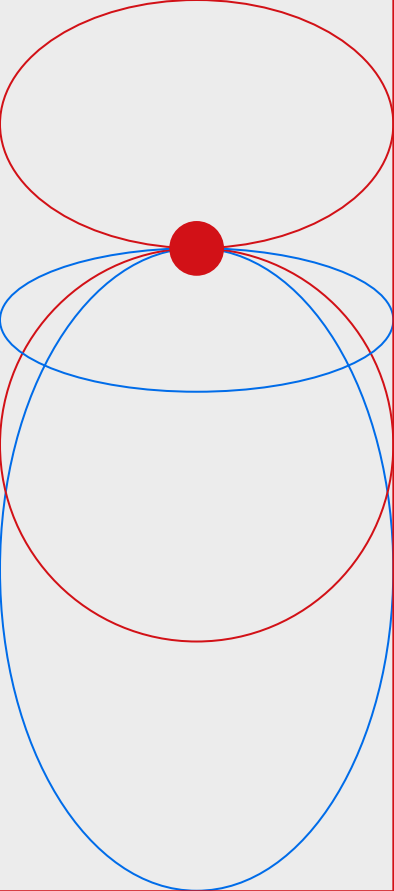
Insight

- High-volume users show a stable “office-hour” usage pattern, while Others peak slightly later in the afternoon.

Recommendation

- Schedule **high-value campaigns/outreach** for high-volume accounts in **09:00–13:00** (especially **11:00**).
- For Others, prioritize **14:00–16:00** to align with their peak engagement.
- Avoid **post-18:00** windows for time-sensitive communications.





Thank you

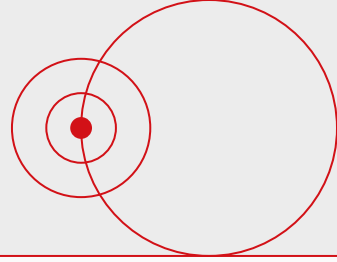
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 karen.amarillys@gmail.com

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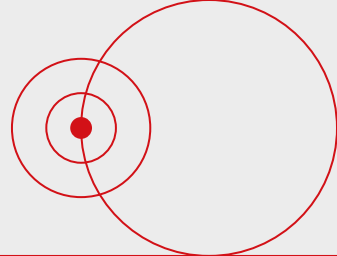


Appendix (Presentation Recording)



[LINK](#)

Appendix (RFM)



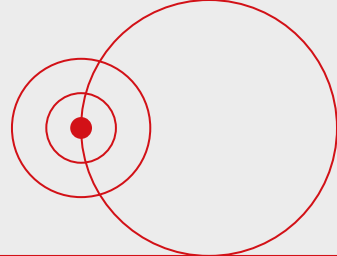
```
WITH interactions AS (  
  SELECT  
    user_id,  
    TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000  
YEAR)) AS event_time,  
    'Call' AS channel,  
    CAST(message_status AS STRING) AS status,  
    is_charge,  
    CAST(total_price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Call`
```

```
  UNION ALL  
  SELECT  
    user_id,  
    TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000  
YEAR)) AS event_time,  
    'Email' AS channel,  
    CAST(message_status AS STRING) AS status,  
    is_charge,  
    CAST(price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Email`
```

```
  UNION ALL  
  SELECT  
    user_id,  
    TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000  
YEAR)) AS event_time,  
    'Sms' AS channel,  
    CAST(message_status AS STRING) AS status,  
    is_charge,  
    CAST(total_price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Sms`
```

```
  UNION ALL  
  SELECT  
    user_id,  
    TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000  
YEAR)) AS event_time,  
    'Whatsapp' AS channel,  
    CAST(last_status AS STRING) AS status,  
    is_charge,  
    CAST(price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Whatsapp),
```

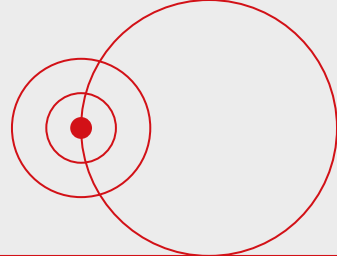

Appendix



```
anchor AS (  
  SELECT DATE(MAX(event_time)) AS anchor_date  
  FROM interactions  
) ,  
  
rfm_raw AS (  
  SELECT  
    user_id,  
    DATE_DIFF((SELECT anchor_date FROM anchor),  
DATE(MAX(event_time)), DAY) AS recency_days,  
    COUNT(DISTINCT DATE(event_time)) AS frequency_active_days,  
    SUM(CASE WHEN is_charge THEN revenue ELSE 0 END) AS  
monetary_revenue,  
  
    COUNT(DISTINCT channel) AS channel_diversity,  
    COUNT(*) AS total_events,  
    SUM(CASE WHEN is_charge THEN 1 ELSE 0 END) AS  
ORDER BY segment, rfm_total DESC, recency_days ASC;
```

```
charged_events,  
  MIN(event_time) AS first_activity_ts,  
  MAX(event_time) AS last_activity_ts  
FROM interactions  
GROUP BY 1  
) ,  
  
with_users AS (  
  SELECT  
    r.*,  
    u.user_name,  
    u.field_of_business,  
    u.join_date,  
    DATE_DIFF((SELECT anchor_date FROM anchor), u.join_date,  
DAY) AS tenure_days  
FROM rfm_raw r  
LEFT JOIN `revou-vi-001.telkom_oca.Users` u  
USING (user_id)  
) ,
```

Appendix

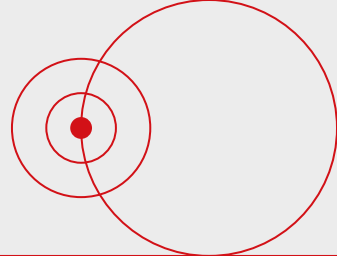


```
bounds AS (  
  SELECT  
    MIN(monetary_revenue) AS min_m,  
    MAX(monetary_revenue) AS max_m  
  FROM with_users  
) ,  
  
scored AS (  
  SELECT  
    w.*,  
  
    -- R (guardrail)  
    CASE  
      WHEN MIN(recency_days) OVER () = MAX(recency_days) OVER ()  
    THEN 5  
      ELSE NTILE(5) OVER (ORDER BY recency_days ASC, user_id)  
    END AS r_score,
```

```
    -- F (guardrail)  
    CASE  
      WHEN MIN(frequency_active_days) OVER () =  
      MAX(frequency_active_days) OVER () THEN 5  
      ELSE NTILE(5) OVER (ORDER BY frequency_active_days DESC,  
user_id)  
    END AS f_score,  
  
    -- M (min-max binning into 1..5)  
    CASE  
      WHEN (SELECT max_m FROM bounds) = (SELECT min_m FROM  
bounds) THEN 5  
      WHEN monetary_revenue = (SELECT max_m FROM bounds) THEN 5  
      ELSE  
        LEAST(  
          5,  
          GREATEST(  
            1,  
            CAST(  
              FLOOR(  
                1 + 5 * SAFE_DIVIDE(  

```

Appendix



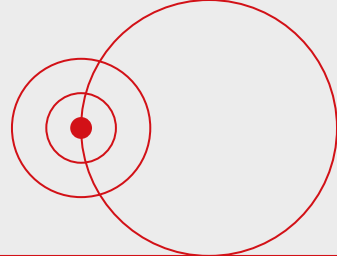
```
        (monetary_revenue - (SELECT min_m FROM
bounds)),
        ((SELECT max_m FROM bounds) - (SELECT min_m
FROM bounds))
    )
    ) AS INT64
    )
    )
    )
    END AS m_score

FROM with_users w
),

final AS (
    SELECT
        s.*,
        (s.r_score + s.f_score + s.m_score) AS rfm_total,
        CONCAT(CAST(s.r_score AS STRING), CAST(s.f_score AS STRING),
CAST(s.m_score AS STRING)) AS rfm_cell,
```

```
    CASE
        WHEN s.join_date IS NOT NULL AND s.tenure_days < 14 THEN
'New (Tenure<14d)'
        WHEN s.r_score >= 4 AND s.f_score >= 4 AND s.m_score >= 4
THEN 'Champions'
        WHEN s.r_score >= 4 AND s.f_score >= 4 THEN 'Loyal'
        WHEN s.r_score >= 4 AND (s.f_score BETWEEN 2 AND 3) THEN
'Potential Loyalists'
        WHEN s.r_score = 5 AND s.f_score = 1 THEN 'New
(Behavior)'
        WHEN s.r_score <= 2 AND (s.f_score >= 4 OR s.m_score >=
4) THEN 'At Risk'
        WHEN s.r_score <= 2 AND s.f_score <= 2 THEN 'Hibernating'
        ELSE 'Needs Attention'
    END AS segment
FROM scored s
)
```

Appendix

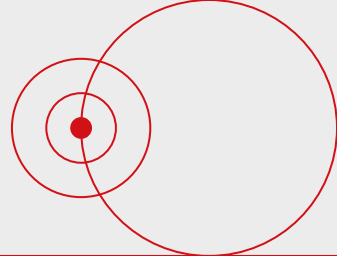


SELECT

```
user_id,  
user_name,  
field_of_business,  
join_date,  
tenure_days,  
recency_days,  
frequency_active_days,  
monetary_revenue,  
channel_diversity,  
total_events,  
charged_events,  
r_score, f_score, m_score,  
rfm_total,  
rfm_cell,  
segment,  
first_activity_ts,  
last_activity_ts
```

FROM final

Appendix (segmentation by monetary)



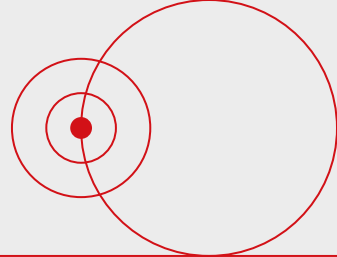
```
WITH interactions AS (  
  SELECT  
    user_id,  
    TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000  
YEAR)) AS event_time,  
    CAST(is_charge AS BOOL) AS is_charge,  
    CAST(total_price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Call`
```

```
  UNION ALL  
  SELECT  
    user_id,  
    TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000  
YEAR)) AS event_time,  
    CAST(is_charge AS BOOL) AS is_charge,  
    CAST(price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Email`
```

```
  UNION ALL  
  SELECT  
    user_id,  
    TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000  
YEAR)) AS event_time,  
    CAST(is_charge AS BOOL) AS is_charge,  
    CAST(total_price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Sms`
```

```
  UNION ALL  
  SELECT  
    user_id,  
    TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000  
YEAR)) AS event_time,  
    CAST(is_charge AS BOOL) AS is_charge,  
    CAST(price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Whatsapp`  
) ,
```

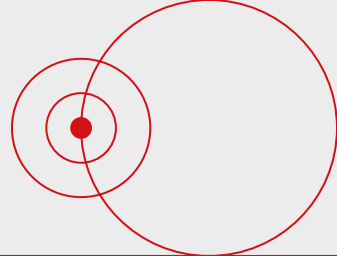
Appendix



```
monetary AS (  
  SELECT  
    user_id,  
    SUM(CASE WHEN is_charge THEN revenue ELSE 0 END) AS  
monetary_revenue  
  FROM interactions  
  GROUP BY 1  
,  
  
with_users AS (  
  SELECT  
    m.user_id,  
    u.user_name,  
    u.field_of_business,  
    m.monetary_revenue  
  FROM monetary m  
  LEFT JOIN `revou-vi-001.telkom_oca.Users` u  
  USING (user_id)  
,
```

```
segmented AS (  
  SELECT  
    w.*,  
    CASE  
      WHEN NTILE(3) OVER (ORDER BY monetary_revenue ASC,  
user_id) = 1 THEN 'Low'  
      WHEN NTILE(3) OVER (ORDER BY monetary_revenue ASC,  
user_id) = 2 THEN 'Medium'  
      ELSE 'High'  
    END AS monetary_segment  
  FROM with_users w  
)  
  
SELECT  
  user_id,  
  user_name,  
  field_of_business,  
  monetary_revenue,  
  monetary_segment  
FROM segmented  
ORDER BY monetary_segment, monetary_revenue DESC;
```

Appendix (Churn Detection)



```
WITH interactions AS (  
  -- CALL  
  SELECT user_id,  
         TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000 YEAR)) AS  
event_time,  
         'Call' AS channel,  
         is_charge,  
         CAST(total_price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Call`  
  
  UNION ALL
```

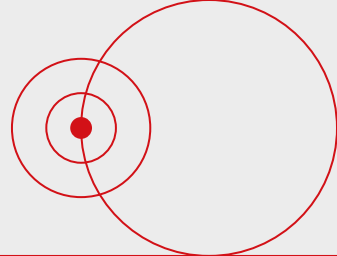
```
  -- EMAIL  
  SELECT user_id,  
         TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000 YEAR)) AS  
event_time,  
         'Email' AS channel,  
         is_charge,  
         CAST(price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Email`  
  
  UNION ALL
```

```
  -- SMS  
  SELECT user_id,  
         TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000 YEAR)) AS  
event_time,  
         'Sms' AS channel,  
         is_charge,  
         CAST(total_price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Sms`  
  
  UNION ALL
```

```
  -- WHATSAPP  
  SELECT user_id,  
         TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000 YEAR)) AS  
event_time,  
         'Whatsapp' AS channel,  
         is_charge,  
         CAST(price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Whatsapp`  
) ,
```

```
anchor AS (  
  SELECT DATE(MAX(event_time)) AS anchor_date  
  FROM interactions  
) ,
```

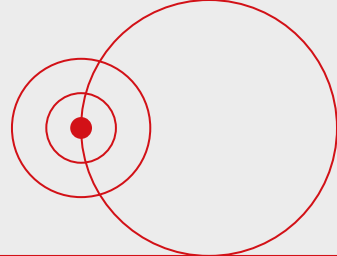
Appendix



```
windowed AS (  
  SELECT  
    i.*,  
    (SELECT anchor_date FROM anchor) AS anchor_date  
  FROM interactions i  
) ,  
  
per_user AS (  
  SELECT  
    user_id,  
  
    -- Activity & value in last 30d vs previous 30d  
    SUM(CASE WHEN DATE(event_time) > DATE_SUB(anchor_date, INTERVAL 30 DAY)  
      AND DATE(event_time) <= anchor_date THEN 1 ELSE 0 END) AS  
events_last_30d,  
    SUM(CASE WHEN DATE(event_time) > DATE_SUB(anchor_date, INTERVAL 60 DAY)  
      AND DATE(event_time) <= DATE_SUB(anchor_date, INTERVAL 30 DAY) THEN  
1 ELSE 0 END) AS events_prev_30d,  
  
    SUM(CASE WHEN is_charge  
      AND DATE(event_time) > DATE_SUB(anchor_date, INTERVAL 30 DAY)  
      AND DATE(event_time) <= anchor_date THEN 1 ELSE 0 END) AS  
charged_events_last_30d,  
    SUM(CASE WHEN is_charge  
      AND DATE(event_time) > DATE_SUB(anchor_date, INTERVAL 60 DAY)  
      AND DATE(event_time) <= DATE_SUB(anchor_date, INTERVAL 30 DAY) THEN  
1 ELSE 0 END) AS charged_events_prev_30d,
```

```
    SUM(CASE WHEN is_charge  
      AND DATE(event_time) > DATE_SUB(anchor_date, INTERVAL 30 DAY)  
      AND DATE(event_time) <= anchor_date THEN revenue ELSE 0 END) AS  
revenue_last_30d,  
    SUM(CASE WHEN is_charge  
      AND DATE(event_time) > DATE_SUB(anchor_date, INTERVAL 60 DAY)  
      AND DATE(event_time) <= DATE_SUB(anchor_date, INTERVAL 30 DAY) THEN  
revenue ELSE 0 END) AS revenue_prev_30d,  
  
    -- Change rates  
    SAFE_DIVIDE(  
      SUM(CASE WHEN is_charge AND DATE(event_time) > DATE_SUB(anchor_date,  
INTERVAL 30 DAY) AND DATE(event_time) <= anchor_date THEN revenue ELSE 0 END)  
      -  
      SUM(CASE WHEN is_charge AND DATE(event_time) > DATE_SUB(anchor_date,  
INTERVAL 60 DAY) AND DATE(event_time) <= DATE_SUB(anchor_date, INTERVAL 30 DAY)  
THEN revenue ELSE 0 END),  
      NULLIF(SUM(CASE WHEN is_charge AND DATE(event_time) > DATE_SUB(anchor_date,  
INTERVAL 60 DAY) AND DATE(event_time) <= DATE_SUB(anchor_date, INTERVAL 30 DAY)  
THEN revenue ELSE 0 END), 0)  
    ) AS revenue_change_pct,
```


Appendix

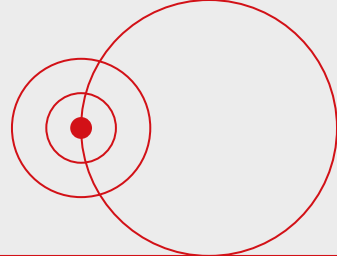


```
SAFE_DIVIDE(  
    SUM(CASE WHEN is_charge AND DATE(event_time) > DATE_SUB(anchor_date,  
INTERVAL 30 DAY) AND DATE(event_time) <= anchor_date THEN 1 ELSE 0 END)  
    -  
    SUM(CASE WHEN is_charge AND DATE(event_time) > DATE_SUB(anchor_date,  
INTERVAL 60 DAY) AND DATE(event_time) <= DATE_SUB(anchor_date, INTERVAL 30 DAY)  
THEN 1 ELSE 0 END),  
    NULLIF(SUM(CASE WHEN is_charge AND DATE(event_time) > DATE_SUB(anchor_date,  
INTERVAL 60 DAY) AND DATE(event_time) <= DATE_SUB(anchor_date, INTERVAL 30 DAY)  
THEN 1 ELSE 0 END), 0)  
    ) AS charged_events_change_pct  
FROM windowed  
GROUP BY user_id  
)
```

```
SELECT  
    u.user_name,  
    p.user_id,          -- keep for traceability (you can remove if you want)  
    p.events_last_30d,  
    p.events_prev_30d,  
    p.charged_events_last_30d,  
    p.charged_events_prev_30d,  
    p.revenue_last_30d,  
    p.revenue_prev_30d,  
    p.revenue_change_pct,  
    p.charged_events_change_pct,
```

```
CASE  
    WHEN p.charged_events_prev_30d >= 20  
        AND SAFE_DIVIDE(p.charged_events_last_30d,  
NULLIF(p.charged_events_prev_30d, 0)) < 0.7  
        THEN 'Churn Risk (Drop ≥30%)'  
        ELSE 'Stable/Normal'  
    END AS churn_flag  
FROM per_user p  
LEFT JOIN `revou-vi-001.telkom_oca.Users` u  
USING (user_id)  
ORDER BY p.revenue_last_30d DESC;
```

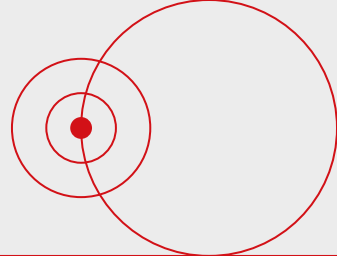
Appendix (Hourly Activities)



```
WITH interactions AS (  
  SELECT user_id,  
         TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000 YEAR)) AS  
event_time,  
         is_charge,  
         CAST(total_price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Call`  
  UNION ALL  
  SELECT user_id,  
         TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000 YEAR)) AS  
event_time,  
         is_charge,  
         CAST(price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Email`  
  UNION ALL  
  SELECT user_id,  
         TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000 YEAR)) AS  
event_time,  
         is_charge,  
         CAST(total_price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Sms`  
  UNION ALL  
  SELECT user_id,  
         TIMESTAMP(DATETIME_ADD(DATETIME(created_at), INTERVAL 2000 YEAR)) AS
```

```
event_time,  
         is_charge,  
         CAST(price AS INT64) AS revenue  
  FROM `revou-vi-001.telkom_oca.Whatsapp`  
) ,  
anchor AS (  
  SELECT DATE(MAX(event_time)) AS anchor_date FROM interactions  
) ,  
  
-- last 30d user value  
user_value AS (  
  SELECT  
    user_id,  
    SUM(CASE WHEN is_charge  
              AND DATE(event_time) > DATE_SUB((SELECT anchor_date FROM anchor),  
INTERVAL 30 DAY)  
              AND DATE(event_time) <= (SELECT anchor_date FROM anchor)  
              THEN revenue ELSE 0 END) AS revenue_last_30d,  
    SUM(CASE WHEN is_charge  
              AND DATE(event_time) > DATE_SUB((SELECT anchor_date FROM anchor),  
INTERVAL 30 DAY)  
              AND DATE(event_time) <= (SELECT anchor_date FROM anchor)  
              THEN 1 ELSE 0 END) AS charged_events_last_30d  
  FROM interactions  
  GROUP BY 1  
) ,
```

Appendix



```
-- top 20% by revenue_last_30d (with 20 users => top 4 users)
ranked AS (
  SELECT
    *,
    NTILE(5) OVER (ORDER BY revenue_last_30d DESC) AS revenue_tile
  FROM user_value
),

grouped AS (
  SELECT
    user_id,
    CASE WHEN revenue_tile = 1 THEN 'High-volume (Top 20%)'
          ELSE 'Others' END AS user_group
  FROM ranked
)

SELECT
  g.user_group,
  EXTRACT(HOUR FROM i.event_time) AS hour_of_day,
  COUNT(*) AS events,
  SUM(CASE WHEN i.is_charge THEN 1 ELSE 0 END) AS charged_events,
  SUM(CASE WHEN i.is_charge THEN i.revenue ELSE 0 END) AS charged_revenue
FROM interactions i
JOIN grouped g USING (user_id)
GROUP BY 1,2
ORDER BY user_group, hour_of_day;
```

```
event_time,
  is_charge,
  CAST(price AS INT64) AS revenue
FROM `revou-vi-001.telkom_oca.Whatsapp`
),
anchor AS (
  SELECT DATE(MAX(event_time)) AS anchor_date FROM interactions
),

-- last 30d user value
user_value AS (
  SELECT
    user_id,
    SUM(CASE WHEN is_charge
              AND DATE(event_time) > DATE_SUB((SELECT anchor_date FROM anchor),
INTERVAL 30 DAY)
              AND DATE(event_time) <= (SELECT anchor_date FROM anchor)
            THEN revenue ELSE 0 END) AS revenue_last_30d,
    SUM(CASE WHEN is_charge
              AND DATE(event_time) > DATE_SUB((SELECT anchor_date FROM anchor),
INTERVAL 30 DAY)
              AND DATE(event_time) <= (SELECT anchor_date FROM anchor)
            THEN 1 ELSE 0 END) AS charged_events_last_30d
  FROM interactions
  GROUP BY 1
),
```